

ROKHAN AHMADZAI

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PROFESSIONAL SUMMARY

Third-year Mechanical Engineering student at RIT seeking a Spring/Summer 2027 co-op in mechanical design or robotics. Hands-on with SolidWorks design, machined and 3D-printed prototypes, and physical load testing; also brings experience in security operations and U.S. military linguistics, with fluency in five languages.

TECHNICAL SKILLS

- **CAD & Analysis:** SolidWorks (parts, assemblies, Simulation), GD&T, Design for Manufacturing (DFM), structural analysis
- **Fabrication & Test:** manual machining, FDM 3D printing, physical load testing, Pasco Capstone (DAQ)
- **Programming & Digital:** Python, C++, MATLAB, VHDL (Quartus Prime, MAX 10 FPGA)
- **Languages:** English (Professional), Dari & Pashto (Native), Urdu & Hindi (Conversational)

EDUCATION

Rochester Institute of Technology (RIT)

Rochester, NY

B.S. in Mechanical Engineering | Expected May 2028 |

Monroe Community College (MCC)

Rochester, NY | Graduated May 2026

A.S. in Engineering Science & Physics

- **Coursework:** Statics, Dynamics, Strength of Materials, MATLAB, Python & Robotics, Digital Systems
- **Honors:** 1st Prize, Design Champion Trophy (Statics); Matt Hachee Award for Engineering Promise

ENGINEERING PROJECTS

CAD-to-Machined Vehicle Chassis | Machining & Prototyping

MCC, 2025

- Primary designer on a two-person team; modeled a competition chassis in SolidWorks and machined/3D-printed it at MCC's Applied Technology Advancement Center. Placed top of class for sprint speed, tow weight, and lap time.

Autonomous Sorting Robot (EV3) | Robotics & Automation

MCC, 2025

- Led Python programming on a three-person team, writing the core navigation and sorting logic; resolved conflicts between the robot's color and light sensors to reliably sort targets within the time limit for the maximum score.

Thunder Wheel | Structural Truss Analysis | Mechanics of Statics

MCC, 2025

- Built a rotating truss and applied Method of Sections to estimate member forces, then compared results against physical load-testing data. Awarded 1st Prize, Design Champion Trophy.

Additional Projects

MCC, 2024–2025

- Digital logic: VHDL-based microwave FSM controller on a DE10-Lite (MAX 10 FPGA) board with BCD 7-segment display; also built a MATLAB physics toolbox app for 3D projectile-motion analysis.

PROFESSIONAL EXPERIENCE

Co-Founder | Wise Web Brothers | Digital Marketing & Media

2025 – 2026

- Built and launched websites for five paid clients (restaurants, car dealerships, mechanic shops); managed social media, content creation, ad campaigns, and branding using Canva, Adobe Premiere, CapCut, and Meta. Deliverables included Google-verified business profiles, map registration, and full-service video and graphic design.

Student Assistant | MCC Libraries

Jan 2024 – May 2026

- Staffed the front desk (patron check-in for study rooms, textbook/calculator/equipment checkout, printing, and navigation help) and assisted with book cataloging, library classroom organization, and public computer upkeep.

Technical Support & Operations (GSOC) | Northland Controls

Jun 2022 – Apr 2023

- Monitored global security infrastructure and executed real-time incident response protocols.

Linguist | U.S. Forces | Afghanistan, Qatar, Germany, USA

2021

- Provided translation and cultural liaison support for multinational military operations across Afghanistan, Qatar, and Germany; service concluded at Joint Base McGuire-Dix-Lakehurst, NJ.

Marketing Supervisor | USAID-Funded Initiative

2018 – 2020

- Directed a regional field team and payroll operations for a USAID-funded outreach initiative.

LEADERSHIP & COMMUNICATION

Public Speaker | MCC & Model UN

2018 – 2026

- Delivered talks on cross-cultural communication and student resilience at MCC and Model UN conferences.

Global Volunteer | UN Cultural Exchange, India

2018

- Partnered with the UN and AIESEC to deliver English-language instruction to underprivileged rural children.

Peer Mentor | TRIO Student Support Services

Jan 2025 – Aug 2025

- Mentored first-generation STEM students on study strategies and progress tracking, 16–20 hrs/week.